

# MACHINE LEARNING CAPSTONE PROJECT

## Model Comparison & Explainability

### Medical Insurance Cost Classification

#### Final Project Report

## 1. Executive Summary

This project develops an end-to-end machine learning classification solution using the Medical Cost Personal Dataset used in a previous internship task. The original dataset is a regression dataset containing individual characteristics and medical insurance charges. For this capstone, the problem was adapted into binary classification by predicting whether a person's medical insurance charge belongs to a High Cost or Low Cost category.

The cleaned dataset contains 1,337 records. The median medical charge, 9386.1613, was used as the classification threshold. Five models were evaluated on the same stratified 80/20 train-test split: Logistic Regression, Random Forest, Decision Tree, XGBoost, and a Stacked Ensemble. The models were compared using Precision, Recall, F1-score, and ROC-AUC, with 5-fold Stratified Cross-Validation used to support final model selection.

XGBoost was selected as the final model using the highest mean F1-score from 5-fold Stratified Cross-Validation (0.9292). On the untouched 20% holdout test set, XGBoost achieved Precision 0.9756, Recall 0.8955, F1-score 0.9339, and ROC-AUC 0.9557. SHAP was used for global and local explainability, while LIME was used for additional local explanations. The final model was saved for reuse and integrated into a Streamlit application with Docker deployment support.

## 2. Project Objective

The objective is to demonstrate a complete machine learning workflow from data inspection and preparation through model comparison, final-model selection, explainability, and deployment. The resulting system predicts whether a medical insurance charge is High Cost or Low Cost from demographic, health-related, and smoking-related features.

Because the source dataset is originally intended for regression, this report explicitly treats the work as a classification adaptation rather than claiming it is an insurance-claims-fraud dataset.

## 3. Dataset Description

The original dataset contains 1,338 records and seven columns: age, sex, BMI, children, smoker, region, and charges. During cleaning, one duplicate record was removed, leaving 1,337 records with no missing values. The first six variables are used as model inputs. The charges column is used only to construct the target and is not supplied to the model.

| Feature  | Type        | Role                     |
|----------|-------------|--------------------------|
| age      | Numerical   | Input                    |
| sex      | Categorical | Input                    |
| bmi      | Numerical   | Input                    |
| children | Numerical   | Input                    |
| smoker   | Categorical | Input                    |
| region   | Categorical | Input                    |
| charges  | Numerical   | Target construction only |

## 4. Data Exploration and Cleaning

Initial inspection was performed to determine dataset dimensions, data types, missing values, duplicates, descriptive statistics, and feature distributions. The dataset was also examined using a medical-charge histogram, categorical frequency counts, numerical summaries, a correlation matrix, and a comparison of medical charges by smoking status.

The notebook inspection found no missing values and one duplicate record. The duplicate was removed during cleaning, reducing the dataset from 1,338 to 1,337 records. Exploratory analysis also showed that smoking status has a strong relationship with medical charges, making it an important input feature for the classification task.

## 5. Target Engineering

The original continuous charges variable was converted into a binary target named High\_Cost using the median medical charge.

**Threshold = 9386.1613**

Records with charges greater than or equal to 9386.1613 receive High\_Cost = 1. Records below the threshold receive High\_Cost = 0.

The charges column is excluded from X after target construction. This prevents target leakage because the target is directly derived from charges.

## 6. Feature Engineering and Preprocessing

The data was divided into numerical and categorical features. Numerical variables (age, bmi, and children) were standardized with StandardScaler. Categorical variables (sex, smoker, and region) were transformed using OneHotEncoder with handle\_unknown='ignore'. A ColumnTransformer combined both preprocessing paths.

The dataset was split into 80% training and 20% testing data using random\_state=42 and stratification on the target. The same split and preprocessing approach were used for all model comparisons to keep the evaluation fair and reproducible.

## 7. Model Comparison

Five classification approaches were trained and evaluated on exactly the same test data. Precision measures the reliability of High Cost predictions, Recall measures how many actual High Cost cases were detected, F1-score balances Precision and Recall, and ROC-AUC measures discrimination across probability thresholds.

| Model               | Precision | Recall | F1     | ROC-AUC |
|---------------------|-----------|--------|--------|---------|
| Logistic Regression | 0.9044    | 0.9179 | 0.9111 | 0.9440  |
| Random Forest       | 0.9528    | 0.9030 | 0.9272 | 0.9539  |
| Decision Tree       | 0.8889    | 0.8955 | 0.8922 | 0.8918  |
| XGBoost             | 0.9756    | 0.8955 | 0.9339 | 0.9557  |
| Stacked Ensemble    | 0.9606    | 0.9104 | 0.9349 | 0.9505  |

## 8. Final Model Selection

XGBoost was selected based on the highest mean F1-score from 5-fold Stratified Cross-Validation: 0.9292. This selection criterion balances Precision and Recall. On the untouched holdout test set, XGBoost achieved Precision 0.9756, Recall 0.8955, F1-score 0.9339, and ROC-AUC 0.9557. Random Forest had the highest mean CV ROC-AUC (0.9543), while the Stacked Ensemble had a slightly higher holdout F1-score (0.9349). These differences are reported explicitly rather than treating one metric as universally best.

The selection includes a trade-off: XGBoost Recall was 0.8955 on the holdout test set, below Logistic Regression (0.9179) and the Stacked Ensemble (0.9104). The selected model therefore reflects the use of mean CV F1 as the primary selection criterion rather than maximizing Recall alone.

The final XGBoost configuration used 200 estimators, maximum tree depth of 4, learning rate of 0.05, and random\_state=42.

## 9. Error Analysis

A confusion matrix was used to examine false positives and false negatives. For the final XGBoost model, the notebook reports 3 false positives and 14 false negatives on the holdout test set. False positives are Low Cost cases predicted as High Cost, while false negatives are High Cost cases predicted as Low Cost.

This distinction is important because model selection should consider the consequences of different types of errors rather than relying only on a single score. The notebook therefore reports both false-positive and false-negative counts for the final model.

## 10. SHAP Explainability

SHAP (SHapley Additive exPlanations) was used to explain the XGBoost model at both global and local levels.

### 10.1 Global Explanation

The global SHAP summary identifies which transformed features have the greatest influence across the test set. Positive SHAP contributions move a prediction toward High Cost, while negative contributions move it toward Low Cost.

### 10.2 Local Explanations

Three individual test cases were examined using local SHAP explanations. These explanations show how the feature values of a specific person contributed to that person's prediction.

The project stores generated SHAP outputs in the explainability directory, including a global summary and three local patient-case visualizations.

## 11. LIME Explainability

LIME was used as a second local explainability method. Three individual cases were explained to demonstrate how the model behaves around specific inputs. LIME provides a local approximation of the model and identifies features that contribute to the individual prediction.

Using both SHAP and LIME provides complementary evidence for interpreting individual model decisions.

## 12. Deployment

The trained preprocessing pipeline and XGBoost classifier were saved as `final_model.pkl`. A reusable prediction function was also tested on sample records.

A Streamlit application was developed to accept age, sex, BMI, number of children, smoking status, and region. The application returns a High Cost or Low Cost prediction, the probability of High Cost, and a SHAP-based local explanation.

Docker was used to package the application together with its Python dependencies. The project includes a `Dockerfile`, `requirements.txt`, and `.dockerignore` to support reproducible deployment.

## 13. Project Deliverables

- Documented Jupyter notebook covering the end-to-end workflow.
- Five-model comparison using Precision, Recall, F1-score, and ROC-AUC.
- Saved final XGBoost model.
- Global and local SHAP explanations.
- Three LIME local explanations.
- Streamlit prediction and explainability application.
- Dockerized application for deployment.
- Model comparison CSV and explainability image outputs.

## 14. Strengths and Limitations

## Strengths

- All five models were evaluated on the same stratified test split.
- Multiple evaluation metrics were used instead of relying only on accuracy.
- The final model combines strong predictive performance with SHAP and LIME explainability. On the holdout test set, XGBoost achieved 0.9756 Precision, 0.8955 Recall, 0.9339 F1-score, and 0.9557 ROC-AUC.
- The preprocessing pipeline is saved together with the model, reducing inconsistencies between training and deployment.
- The project includes a working Streamlit interface and Docker packaging.

## Limitations

- The source dataset is relatively small, with 1,338 original records and 1,337 records after duplicate removal.
- The High Cost target is artificially constructed from the median charge because the original dataset is a regression dataset.
- The target threshold does not represent a clinical or insurance-industry standard.
- The model should not be interpreted as a clinical, financial, or insurance decision-making system without additional domain validation and real-world data.

## 15. Conclusion

This project demonstrates a complete machine learning classification workflow using a real-world medical insurance cost dataset. After exploratory analysis, cleaning, preprocessing, and feature engineering, five classification models were compared under consistent evaluation conditions and further assessed using 5-fold Stratified Cross-Validation.

XGBoost was selected using the highest mean CV F1-score of 0.9292. On the untouched holdout test set it achieved Precision 0.9756, Recall 0.8955, F1-score 0.9339, and ROC-AUC 0.9557. SHAP and LIME were then used to make the model's predictions more interpretable at global and individual levels.

The final system extends beyond model training by providing a reusable saved model, a Streamlit prediction application, and Docker-based deployment packaging. Overall, the project demonstrates the complete path from raw data to an explainable and deployable machine learning product.